

Work group project

Template for solution



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Table of Contents

Table of Content	Error! Bookmark not defined.
1. Group Information	3
2. Solution Summary	3
3. Process	4
3.1 Design Brief	5
3.1.1 Problem Description	5
3.1.2 Stakeholder Analysis	6
3.3.3 Abstract of the Solution	8
3.3.4 Objectives of the Solution	8
3.2 Solution Development	9
3.2.1 Technology Solution Identification.....	9
3.2.1 Process Solution Identification	11
3.2.3 Proposed Solution Description.....	13
3.3 Solution Assessment	14
3.3.1 Assessment Framework.....	14
3.3.2 Assessment Results	15
4. Discussion of the Process	17
4.1 Use of Artificial Intelligence.....	17
4.2 Sustainability Impact	18
5. Feasibility & Next Steps	19
5.1 Process Feasibility	19
5.2 Technical Feasibility	19
5.3 Next Steps	20

1. Group Information

Team name: BrotherInnovation

University: UNIVERSITY OF PISA

Members (name and study program): HOWLADER SAGOR, AMIN AL MD

Selected Organization: DISCENTIS.IT

Challenge Name: VOCATIONAL LANGUAGE GAP

2. Solution Summary

Title of the solution: TECHBRIDGE AI: FACILITATING VOCATIONAL LANGUAGE ACCESS FOR MIGRANT WORKERS

Objective of the solution: The objective of TechBridge AI is to reduce the vocational language barrier that prevents skilled migrant workers in Italy from accessing qualified employment. The platform aims to improve technical language comprehension and increase professional certification success rates, supporting the transition of migrants from unskilled to skilled occupations within vocational training programs.

Short description (abstract): TechBridge AI is a digital learning platform developed in collaboration with Discentis to support vocational education for migrant workers. The solution addresses the limitation of generic language-learning tools by focusing on **trade-specific technical Italian**. The platform integrates three core functionalities: (1) An AI-generated technical glossary that extracts and simplifies terminology from professional manuals; (2) AI-based role-play simulations that reproduce real workplace communication scenarios; and (3) A skills-alignment module that connects learners' existing professional backgrounds with local certification requirements. By acting as a learning facilitator, TechBridge AI enhances inclusion, reduces skills underutilization, and contributes to SDG 4 (Quality Education) and SDG 8 (Decent Work)

Technical description: TechBridge AI is built on a Python-based backend with an API-first architecture to ensure integration with existing Discentis learning systems. Large Language Models (LLMs) are used to analyze vocational training materials, identify domain-specific terminology, and generate simplified Italian explanations alongside native-language translations. Natural Language Processing techniques support terminology extraction and semantic alignment across languages. Visual learning components are integrated through image retrieval or generation models to associate technical terms with diagrams and tools. The platform is designed to be mobile-first, enabling use in training centers, workshops, and job-site environments.

Link to the presentation of the solution:

Related resources (technical document, python script, r code script, prototyping...):

- Final Project Report (PDF): **TechBridge_Final_Report.pdf**
- Prototype Mock-ups: AI Technical Glossary Interface, Workplace Role-Play Simulator
- Stakeholder Analysis: Interviews with vocational instructors and migrant support organizations
- Data Sources: *ISTAT Migrant Employment Reports 2024*, *Italian Economic Journal 2024*,

3. Process

The design process followed a structured and iterative approach, progressing from problem analysis to solution validation, with continuous alignment to stakeholder needs and sustainability goals.

Phase 1: Problem Identification and Analysis

The project began with an analysis of labor market and vocational education studies highlighting the underutilization of skilled migrant workers in Italy due to language barriers. Reports from ISTAT and CEDEFOP indicate that many migrants possess professional qualifications but are employed in low-skilled roles. Academic literature on migration and education further identifies technical language proficiency as a critical barrier to certification and employment. Initial discussions with vocational educators and migrant support organizations confirmed three recurring issues: lack of trade-specific language resources, difficulty passing technical exams despite existing skills, and excessive workload for instructors required to adapt materials manually.

Phase 2: Stakeholder and Needs Assessment

Three primary stakeholder groups were identified. Migrant workers were defined as the main users, requiring accessible and profession-specific language support. Vocational instructors represented secondary users, needing scalable tools to support learners across multiple trades. Discentis was identified as the institutional partner, seeking innovative, practical solutions aligned with its mission to empower educators and improve learning outcomes.

Phase 3: Solution Design and Technology Selection

Multiple solution approaches were explored, including traditional language courses, static digital glossaries, and manual tutoring models. Based on feasibility, scalability, cost-efficiency, and pedagogical impact, an AI-assisted approach was selected. The final design integrates three core components: an AI-generated technical glossary, AI-driven workplace role-play simulations, and a skills-alignment module connecting prior experience with certification requirements. Large Language Models and Natural Language Processing techniques were chosen as the most suitable technologies to support this design.

Phase 4: Process and Workflow Development

Key user workflows were defined, including learner onboarding, manual upload by instructors, automated glossary generation, interactive role-play sessions, and progress tracking. Clear responsibilities were assigned to each actor involved: learners, instructors, Discentis administrators, and platform operators. The workflows were designed to minimize instructor effort while maximizing learner autonomy and engagement.

Phase 5: Prototyping and Validation

To validate the concept before full development, a prototyping approach was adopted. Low-fidelity interactive mockups were created for the AI Technical Glossary Interface and the Workplace Role-Play Simulator. These prototypes were used to test core assumptions, such as learner engagement with technical vocabulary and usability in vocational settings. This approach allowed early validation through observed user behavior rather than assumptions alone.

Phase 6: Solution Assessment

The proposed solution was evaluated across three dimensions: Technical Feasibility (availability of AI technologies, data accessibility, scalability), Operational Viability (ease of integration into Discentis programs, instructor adoption, cost sustainability), and stakeholder alignment (relevance to learner needs and educational objectives). The assessment confirmed that TechBridge AI is feasible with current technologies, with identified risks such as language complexity variation addressed through iterative refinement.

3.1 Design Brief

3.1.1 Problem Description

Core Issue

Italy currently lacks an accessible and scalable system to help skilled migrant workers bridge the gap between their professional skills and employment opportunities. While migrants often have qualifications in fields like electrical work, plumbing, healthcare, and mechanics, technical language barriers prevent them from passing certification exams and entering their trained professions. This leads to widespread underemployment, or “brain waste,” where skilled workers are forced into low-skilled jobs.

The Main Challenge

The primary challenge is that generic language-learning tools fail to provide trade-specific vocabulary, and vocational instructors cannot manually create individualized materials for learners across diverse professions. This mismatch between skills and job market requirements creates both social and economic inefficiencies, limiting the migrants' career opportunities and leaving employers without qualified personnel.

Context (Industry, Market, Social Setting)

This problem exists at the intersection of vocational education, workforce integration, and social inclusion. Italy faces a shortage of skilled workers in key sectors, while migrants constitute a growing part of the labor supply. Discentis, as a vocational training provider, operates in this context and seeks to support instructors and learners. However, the lack of a system to provide trade-specific language support and track learning outcomes prevents effective integration of skilled migrants into the local job market. The issue aligns with SDG 4 (Quality Education) and SDG 8 (Decent Work and Economic Growth).

Causes and Impacts

Causes

- Generic language apps and courses teach everyday language, not vocational terminology.
- Certification exams require technical Italian vocabulary that migrants often lack.
- Instructors are overloaded, often teaching multiple trades, and cannot prepare customized materials for each learner.
- No standardized system exists to map prior skills to local certification requirements.

Impacts

- Skilled migrants remain underemployed in unskilled jobs.

- Employers cannot fill positions with qualified personnel, worsening labor shortages.
- Vocational instructors experience burnout trying to adapt materials for multiple trades.
- Migrants' professional confidence and social inclusion are diminished.
- The national workforce suffers economic inefficiency due to underutilized skills.

3.1.2 Stakeholder Analysis

Primary Users: Migrant Workers

Migrant workers are the central users of TechBridge AI. They include skilled individuals in trades such as electricians, plumbers, healthcare workers, and mechanics who have recently moved to Italy. While they possess professional experience, they face language barriers that prevent them from passing technical certification exams and accessing jobs in their trained professions.

Their needs include:

- Access to trade-specific technical Italian rather than generic language learning.
- AI-assisted tools that provide visual, bilingual, and contextual explanations of professional terms.
- Opportunities to practice workplace communication in simulated scenarios.
- Guidance to map their existing skills to local certification requirements.
- Flexibility to learn at their own pace, remotely or on-site.

These users value practical, immediately applicable learning that enables them to work confidently in their profession and integrate socially and economically in Italy.

Secondary Users: Vocational Instructors

Vocational instructors are secondary users. They teach migrants across multiple trades but cannot manually create individualized resources for each learner. Their current challenges include:

- Overload from needing to teach multiple trades simultaneously.
- Lack of tools to provide customized, trade-specific language support.
- Difficulty monitoring learner progress across language and skill development.

By using TechBridge AI, instructors gain a scalable way to support learners without creating resources manually, freeing them to focus on pedagogical guidance and mentoring.

Institutional Partner: Discentis

Discentis plays a key role in providing vocational training and facilitating migrant workforce integration. Their needs include:

- Tools to reach learners across Italy, including underserved regions.
- Mechanisms to track learner progress and identify gaps in technical language understanding.
- Support for instructors to deliver personalized, trade-specific content efficiently.
- Insights into which training elements improve exam pass rates and employment outcomes.

Constraints and Limitations

Migrant Workers:

- Limited proficiency in technical Italian prevents them from passing certification exams.
- Busy schedules and economic pressures reduce time for additional study.
- Existing language apps focus on general vocabulary, not vocational terminology.

Instructors:

- Cannot manually adapt teaching materials for each profession.
- Manage students across diverse trades, increasing workload.

Discentis:

- Cannot track learner language-specific gaps.
- Limited insight into which learners succeed or drop out.
- Current tools do not support scalable, trade-specific training.

Identification Process

We used a multi-method approach to identify stakeholders and their needs:

1. Quantitative Data Analysis:

- Reviewed ISTAT Migrant Labor Reports and CEDEFOP Vocational Education trends to identify underutilization of skilled migrant labor and certification pass rates.

2. Academic Literature Review:

- Studied research on migration and workforce integration to understand barriers caused by technical language gaps and underemployment.

3. Stakeholder Interviews:

- Conducted interviews with vocational instructors and migrant support NGOs to validate real-world constraints, such as lack of trade-specific resources and overburdened instructors.

4. Stakeholder Mapping:

- Identified primary users (migrant workers), secondary users (instructors), and institutional partners (Discentis).
- Documented their needs, behaviors, values, and constraints.

5. AI Assistance in Synthesis:

- AI tools were used to process large datasets and reports, summarize findings, and identify key patterns. All AI-generated insights were verified by the team.

Data Sources

- ISTAT Migrant Labor Market Reports
- CEDEFOP Vocational Education Trends
- Discentis Internal Training Materials
- Stakeholder Interviews (Vocational Instructors & NG

3.3.3 Abstract of the Solution

What is the General Approach?

TechBridge AI, developed in collaboration with Discentis, is a digital platform that helps skilled migrant workers overcome technical language barriers in Italy. The platform leverages AI-assisted tools to generate trade-specific glossaries, simulate workplace interactions, and match learners' existing skills to local certification requirements. It is designed to support instructors and learners, enhancing vocational training programs without replacing existing resources.

What Value Does It Aim to Deliver?

TechBridge AI delivers three main benefits:

1. **AI-Generated Technical Glossaries:** Automatically extracts trade-specific terminology from manuals, provides translations, and attaches visual aids to accelerate comprehension.
2. **Simulated Workplace Role-Play:** Learners practice real job scenarios with AI acting as supervisors or customers, increasing confidence for oral and practical certification exams.
3. **Skills Alignment Dashboard:** Connects prior professional experience with local certification requirements and tracks learner progress, ensuring efficient workforce integration.

These features directly tackle the key barriers: lack of specialized language support for migrants, high instructor workload, and low certification success rates.

How Does It Connect to the Problem and Users?

- **Primary Users (Migrant Workers):** Overcomes technical language barriers, provides flexible, self-paced learning, and boosts confidence for certifications and job placement.
- **Secondary Users (Instructors):** Reduces manual preparation of trade-specific materials, provides insights into learner progress, and facilitates scalable teaching support.
- **Institutional Stakeholder (Discentis):** Gains tools to track learner outcomes, monitor language proficiency gaps, and enhance training effectiveness.
- **Policy Relevance:** Supports SDG 4 (Quality Education) and SDG 8 (Decent Work) by improving migrant access to skill-aligned employment.

3.3.4 Objectives of the Solution

Functional Objectives

- Provide easy access to AI-generated technical glossaries and workplace simulations from any device.
- Offer personalized support for each trade, including native-language translations and visual aids.
- Ensure learners can practice certification-relevant scenarios and monitor progress.
- Reduce instructor workload by automating glossary creation and providing learner insights.

Economic Objectives

- Optimize the efficiency of vocational training programs by centralizing technical language resources.
- Reduce costs for both learners and Discentis by minimizing the need for printed materials and manual instruction.
- Enable a scalable platform that supports more learners without proportional increases in staff or resources.

Social Sustainability Objectives

- Increase inclusion by giving migrant workers equal access to trade-specific technical language support.
- Provide a digital learning community that connects learners and instructors, mitigating isolation.
- Improve wellbeing and confidence by offering flexible, self-paced learning tailored to individual skills and schedules.

Environmental Sustainability Objectives

- Shift training from in-person to AI-powered digital learning, reducing travel and associated carbon emissions.
- Minimize reliance on printed materials by using digital glossaries, diagrams, and simulations.
- Encourage eco-conscious learning practices through the platform's design and content delivery.

3.2 Solution Development

3.2.1 Technology Solution Identification

Exploration of Possible Technologies

During the solution design phase, we evaluated several technological approaches to determine which could best address migrant workers' language barriers and support vocational instructors:

- **National Vocational Marketplace:**
A platform aggregating courses from Discentis and other training providers. While comprehensive, this was rejected because it could compete with Discentis's existing programs and complicate integration with current training workflows.
- **AI-Powered Technical Glossary Generator:**
An AI system that automatically extracts trade-specific terms from manuals, translates them into the learner's native language, and adds visual aids (diagrams/photos). This approach personalizes learning, reduces preparation time for instructors, and directly addresses the barrier of missing technical vocabulary.
- **Simulated Workplace Role-Play:**
Interactive AI scenarios where learners practice communication with supervisors or customers. While engaging, implementing full VR simulations was deemed too costly and hardware-intensive for initial deployment, so we opted for mobile and desktop AI-driven simulations instead.

- **Community and Peer Support Platform:**

A forum for learner-to-learner and learner-to-instructor interactions. This addresses the isolation many migrants face and encourages structured mentorship. Its success depends on AI-driven engagement and gamification to stimulate participation.

Selection Criteria

Technologies were evaluated based on five key criteria:

- **Feasibility:** Can it be implemented within 6–12 months by a small development team (2–3 engineers, 1 ML specialist)?
- **Cost:** Must fit a €50–80K Year-1 budget suitable for an educational/social impact project.
- **Accessibility:** Should work on mobile devices and low-bandwidth networks to reach users nationwide.
- **Sustainability:** Must scale from hundreds to thousands of learners without major redesign.
- **Stakeholder Alignment:** Must address confirmed gaps: missing technical language support, high instructor workload, and low certification success rates.

Final Technology Choice

Based on these criteria, TechBridge AI integrates four core technologies:

1. AI Technical Glossary Generator (Primary):

- Uses NLP to extract terms from vocational manuals and translate them into learners' native languages.
- Adds visual diagrams/photos for each term to improve retention.
- Learners complete a 5-minute profile (trade, prior experience, certification goals), and the system generates a customized glossary.
- Technical stack: Python (NLP libraries), PostgreSQL for term storage, and optional image-generation models for diagrams.

2. Skills-to-Certification Dashboard (Primary):

- Shows learners' progress in mastering technical vocabulary and readiness for local certification exams.
- Aggregates anonymized data to highlight common gaps and provide instructors with actionable insights.
- Technical stack: Data aggregation, interactive visualizations dashboard.

3. Simulated Workplace Role-Play (Secondary):

- AI-driven scenarios where learners practice responding to typical job situations (e.g., fixing a boiler, explaining a medical device).
- Uses Large Language Models with Retrieval-Augmented Generation for accuracy based on trade manuals.
- Technical stack: OpenAI GPT API or open-source LLM, text-to-speech and speech-to-text APIs for oral practice.

4. Community Engagement Platform (Secondary):

- Moderated forum for peer interaction, resource sharing, and mentorship.
- AI features suggest discussion topics, provide notifications, and gamify participation with badges and reputation points.

- Technical stack: Discourse or custom forum software, AI recommendation engine, notification system.

3.2.1 Process Solution Identification

TechBridge AI operates through six interconnected workflows, transforming the fragmented, skill-to-market pathway for migrant workers into a unified, accessible, and scalable vocational language ecosystem.

Workflow 1 – Learner Discovery and Onboarding

Migrant workers learn about TechBridge AI through vocational training centers, NGOs, employer referrals, or community campaigns.

- Learners register via email or Google Single Sign-On and complete a 5-minute profile: trade, prior experience, certification goals, language proficiency, and preferred learning format.
- Learners receive a personalized dashboard displaying the top 5–10 AI-generated technical glossaries and role-play scenarios, with match scores and explanations.

Workflow 2 – AI-Powered Technical Glossary Generation

Triggered upon onboarding or when new manuals/certification content are uploaded.

- AI extracts trade-specific terms from manuals or PDFs.
- Terms are translated into learners' native languages and paired with visual aids (diagrams, photos, 3D models).
- Learners access interactive glossaries, searchable by term, function, or image.
- Continuous updates occur as learners use the platform, ensuring glossaries evolve with certification requirements.

Workflow 3 – Simulated Workplace Role-Play

Learners practice oral and written communication in AI-generated real-world scenarios:

- AI acts as a foreman, customer, or patient, giving tasks (e.g., "Fix the boiler leak," "Explain the electrical panel issue").
- Learners respond using correct technical Italian terms.
- Large Language Models generate dynamic feedback, while voice recognition enables oral practice.
- Performance data feeds back to the dashboard, highlighting skills mastery and certification readiness.

Workflow 4 – Learner Progress and Certification Tracking

The system tracks each learner's mastery of technical terms and role-play performance:

- Interactive dashboards visualize certification readiness, vocabulary coverage, and scenario completion.
- Alerts notify instructors or mentors if a learner is falling behind.
- Aggregated anonymized data enables stakeholders to monitor regional skill gaps.

Workflow 5 – Community and Mentorship Platform

- Moderated forum connecting learners with peers and instructors:
- Users post questions, share resources, and discuss trade-specific challenges.
- AI suggests discussion topics, notifications, and gamification badges/reputation scores to encourage engagement.
- Mentorship features allow skilled migrants to guide newcomers, fostering peer learning and reducing isolation.

Workflow 6 – AI Facilitator / Support Companion

- Accessible from any page, the AI companion provides instant support:
- Answers questions about tools, procedures, and exam preparation.
- Uses NLP and Large Language Models to provide contextually accurate, trade-specific responses.
- Escalates queries involving safety, legal, or certification issues to human moderators.
- Learners provide feedback on responses to continuously improve AI quality.

Actors and Roles

Primary Users – Migrant Learners:

- Complete profiles, access glossaries, participate in simulations, track progress, and engage in community.
- Provide accurate information and actively use learning materials.
- Target: 1,000+ learners Year 1, scalable to 10,000+.

Instructors / Vocational Trainers (Discentis Partners):

- Upload manuals, oversee learner progress, moderate simulations, and validate glossaries.
- Benefits include reduced manual preparation, enhanced learner outcomes, and improved certification success rates.

TechBridge AI Platform (Technology Operator):

- Generates AI glossaries, runs simulations, tracks progress, moderate community, and provides analytics.
- Responsibilities: 99.5% uptime, GDPR compliance, high AI recommendation accuracy (>85%), and secure data handling.

Policy / Employer Stakeholders:

- Monitor regional integration, skill coverage, and certification outcomes.
- Provide funding support, endorse platform adoption, and use aggregated insights for workforce planning.

Payment Processors (Stripe or equivalents – optional):

- Securely handle paid course access or premium content subscriptions.

3.2.3 Proposed Solution Description

TechBridge AI is a digital platform designed to reduce language barriers for migrant workers seeking to enter their skilled professions in Italy. The platform provides personalized technical vocabulary learning, simulated workplace role-plays, mentorship, and real-time progress tracking. Workers access a dashboard displaying their learning progress, matched technical glossaries, and role-play scenarios aligned with their trade and certification goals. Administrators and partner organizations use analytics dashboards to monitor participation, identify underserved regions, and implement targeted interventions. The AI companion provides instant feedback, practical guidance, and links to relevant materials.

Key Components

Technological:

- **AI Technical Glossary Engine:** Extracts technical terms from manuals and provides translations, visual aids, and usage examples (Python, NLP models).
- **Simulated Workplace Engine:** Interactive role-play scenarios powered by Large Language Models (OpenAI GPT API).
- **Community & Mentorship Platform:** AI-enhanced forum encouraging peer-to-peer support and guidance.
- **Progress & Equity Dashboard:** Tracks learner completion, regional participation, and technical skill coverage.
- **Optional Payment Integration:** For premium courses or certifications (Stripe API).

Procedural:

Six interconnected workflows:

- Learner onboarding and needs assessment (5-minute trade/language profile).
- AI-generated technical glossary creation and updates.
- Simulated workplace role-play with AI feedback.
- Progress tracking and certification readiness monitoring.
- AI companion support (instant, context-aware responses).
- Community engagement and peer mentorship with gamified notifications.

Human:

- **Migrant Workers:** Provide profile data, engage with glossaries, simulations, and community.
- **Trainers / Vocational Instructors:** Upload manuals, validate glossaries, moderate role-play and mentorship.
- **TechBridge AI Operators:** Ensure platform uptime (>99.5%), AI accuracy, and privacy compliance.
- **Policy / Employer Stakeholders:** Monitor regional equity and workforce skill coverage.
- **Human Moderators:** Review flagged AI responses and community posts within 24 hours.

Resources Required

Development: 2–3 web developers, 1 ML/NLP engineer over 6–12 months.

Infrastructure: Standard web stack, API integrations (OpenAI GPT, Stripe).

Data: Learner profiles (trade, experience, certification goals, language level), training manual content, historical usage data, optional postal codes for equity tracking.

Skills: Web development, AI/NLP expertise, technical content moderation in Italian and native languages.

Organizational: Partnerships with vocational training centers and NGOs, endorsement from labor/employment agencies, GDPR-compliant privacy infrastructure.

Objectives Fulfillment

Equitable Access: AI Technical Glossary Engine and Simulated Workplace provide trade-specific language support for workers in all regions, reducing barriers due to language or geographic isolation.

Digital & AI Skills: AI companion provides 24/7 guidance on using technical tools, interpreting manuals, and preparing for oral certification exams.

Hands-on Training & Practice: Simulated workplace scenarios allow learners to practice real job tasks safely, addressing the gap between existing skills and certification requirements.

Peer Support & Mentorship: AI-enhanced community fosters structured peer guidance, replacing reliance on informal or unavailable mentors.

Organizational Feasibility: Partnership with training providers ensures TechBridge AI complements existing programs without competing, while administrators gain actionable insights into regional skill gaps.

This integrated technology and workflow design directly addresses the three main confirmed gaps: language barriers in trade-specific skills, isolation of migrant workers, and preparation for technical certification exams.

3.3 Solution Assessment

3.3.1 Assessment Framework

TechBridge AI was evaluated across three dimensions: technical feasibility, operational viability, and alignment with stakeholder needs. Each dimension uses measurable criteria grounded in realistic conditions for migrant workers, training partners, and vocational institutions.

1. Technical Feasibility

Metrics:

- **Technology maturity:** The platform relies on established technologies including Python-based NLP for glossary extraction, OpenAI GPT API for role-play simulations, and standard web stack tools for dashboards and community forums.

- **Implementation complexity:** A small team (2–3 developers + 1 ML/NLP engineer) can build the platform in 6–12 months using off-the-shelf components and existing AI models.
- **Data availability:** Sufficient data exists from vocational manuals, prior training materials, and worker profiles to support AI recommendations and analytics. Optional postal codes provide regional equity insights.
- **Scalability:** The system is designed to scale from a pilot of 500–1,000 workers to 50,000+ users without fundamental redesign, using cloud hosting and modular AI components.

2. Operational Viability

Metrics:

- **Organizational feasibility:** Training partners, vocational centers, and NGOs are willing and able to integrate TechBridge AI into existing programs. Partnerships provide access to manuals, certification requirements, and participant recruitment.
- **User adoption potential:** Migrant workers are highly motivated to reduce language barriers and prepare for certification exams. Anticipated adoption is high (>60%), supported by personalized AI guidance, gamified learning, and accessible mobile/web delivery.

3. Alignment with Stakeholder Needs

Metrics:

- **Problem coverage:** TechBridge AI addresses all confirmed gaps:
 - Language barriers in technical skills – through the AI-powered technical glossary.
 - Isolation and lack of mentorship – through AI-enhanced community and peer matching.
 - Certification readiness and skill application – via simulated workplace role-plays and AI guidance.
- **Impact on priority need (Skill-to-Market Readiness):** The solution aims to reduce failure rates in certification exams and increase language proficiency for job readiness by $\geq 20\%$ within 18 months.

3.3.2 Assessment Results

Technical Feasibility: Strong

The technologies required for TechBridge AI—AI-based technical language recommendations, interactive dashboards, LLM-powered role-play assistant, and standard web/mobile platforms—are mature and widely used in vocational training and e-learning environments. Similar approaches are successfully deployed by platforms such as Coursera, LinkedIn Learning, and Khan Academy. A small development team (2–3 developers + 1 ML engineer) can realistically implement the platform within 6–12 months, and hosting and AI service costs remain manageable for vocational training organizations or NGOs. Initial AI recommendations can start with content-based matching and gradually improve as more migrant worker interactions generate data.

Operational Viability: Moderate to Strong

TechBridge AI aligns with stakeholder-confirmed gaps: lack of accessible technical language training, isolation in learning, and difficulty in certification readiness. Organizational support from partner NGOs and training centers is strong. Worker adoption is expected to be moderate to high, as nearly all participants face barriers in accessing formal technical language resources. Sustaining engagement over time will require active measures, such as gamified prompts, reminders, and peer interactions. Platform maintenance is feasible with one full-time operator and part-time moderators for content and AI oversight.

Alignment with Stakeholder Needs: Very Strong

The solution addresses all identified needs:

- **Primary users (migrant workers):** Receive personalized AI-guided technical language training, 24/7 role-play support, and peer interaction opportunities.
- **Training providers/partners:** Gain actionable data on regional participation, learner progress, and engagement patterns.
- **Government/Policy stakeholders:** Access real-time equity monitoring and evidence-based insights to guide resource allocation and measure SDG 4 alignment.

TechBridge AI ensures direct impact on the priority challenge: reducing technical language barriers that limit access to vocational certifications and labor market opportunities.

Key Strengths

- **Proven technology:** Uses established AI methods, dashboards, and LLMs, reducing development risk and supporting a 6–12-month rollout.
- **Directly aligned with validated stakeholder problems:** Solves the three main gaps confirmed by partners:
 1. Language barriers in vocational skills
 2. Learner isolation and lack of mentorship
 3. Low readiness for certification
- **Comprehensive coverage of needs:** Integrated features include personalized AI training, peer learning, role-play simulations, and real-time equity monitoring.
- **Clear value for stakeholders:**
 - Workers: faster skill acquisition, personalized guidance, 24/7 support
 - Partners: data insights, improved training effectiveness
 - Policy stakeholders: actionable equity and participation data
- **Scalable design:** Modular cloud architecture allows growth from pilot (500–1,000 users) to 50,000+ users without redesign.

Key Weaknesses and Mitigation Strategies

- **Sustaining engagement:** Initial adoption is likely high due to motivation and novelty, but ongoing activity may decline. Mitigation includes gamification, progress badges, and regular AI-driven prompts.
- **Dependence on partner collaboration:** The platform's effectiveness relies on consistent input

from NGOs/training providers (updating resources, responding to enrollment data, providing completion tracking). Mitigation includes formal agreements and automated reminders.

Potential Improvements (Future Iterations)

- Multilingual support for diverse migrant populations
- Offline access for low-connectivity regions
- Integration with existing LMS platforms such as Moodle, DTLS, or Microsoft Teams

4. Discussion of the Process

Social Impact:

TechBridge AI directly addresses inequities in access to professional language training for migrant workers. By providing personalized, AI-powered learning paths, the platform ensures that participants from underserved regions receive the guidance and support they need. Peer interaction features and the AI assistant create opportunities for mentorship, collaboration, and skill practice, helping users build confidence and social connections. Overall, the platform strengthens inclusion, empowers learners, and reduces barriers that previously limited participation.

Environmental Impact:

Shifting from traditional in-person training to a primarily digital platform significantly reduces the environmental footprint. Participants no longer need to travel to physical training centers, which lowers transportation-related carbon emissions. Additionally, the platform minimizes the use of printed materials, relying instead on digital resources. While the server infrastructure does consume energy, selecting green hosting solutions can offset most of the environmental impact, making the system a sustainable alternative to conventional training programs.

Economic Impact:

The platform optimizes both organizational and individual efficiency. For providers, centralized course delivery and automated AI recommendations reduce administrative overhead, lower dropout rates, and improve engagement without significantly increasing costs. Participants benefit economically as well, since they save time searching for suitable courses and can access affordable or free training, increasing their employability and career opportunities. Over time, the platform creates measurable economic value for both the institution and its users.

4.1 Use of Artificial Intelligence

How AI was used

AI was employed in two main ways. First, it served as a research and methodological tool, helping us generate a simulated dataset based on the ISTAT Migrant Employment report, classify. Second, AI was integrated into the solution itself, powering the recommendation engine, the equity dashboard, the AI teaching companion, and smart notifications for the community platform.

Usefulness and quality

AI proved highly useful, saving time in data generation, analysis, and prototyping. The outputs were

accurate enough to support our hypotheses, but initial iterations required refinement. Early datasets lacked detailed regional classification and sufficient records, demonstrating that AI relies heavily on well-defined prompts and human guidance for high-quality outputs.

How usage could improve

AI usage could have been more effective with clearer initial prompts and by feeding AI intermediate outputs to guide subsequent steps. This would reduce repeated iterations and improve the quality of the generated data or suggestions in fewer attempts. Combining multiple AI tools or data sources could also enhance reliability and completeness.

Overall impact

AI significantly influenced the project, enabling evidence-based validation of our problem, supporting statistical testing, and accelerating prototyping. Within EquiLearn, AI became a core feature, personalizing teacher recommendations, monitoring equity, providing 24/7 pedagogical support, and fostering peer interaction. Its contribution strengthened both analysis and implementation, making it central to the final solution.

4.2 Sustainability Impact

Environmental Sustainability

The process we followed minimized environmental impact by reducing the need for extensive physical surveys and campaigns. Traditional methods would have required additional staff, travel, and resources to collect representative samples and validate participation inequalities across Northern and Southern regions. By leveraging AI and simulated datasets, we achieved accurate results without the environmental cost associated with conventional field data collection.

Technological Sustainability

Our approach is designed to integrate seamlessly with Discentis's existing systems, avoiding a complete rebuild of their website or IT infrastructure. Simple backend modifications allow the platform and its features to function effectively, ensuring long-term maintainability and scalability without excessive technical overhead.

Social Sustainability

The solution promotes equity by enabling every teacher to access Discentis's training programs, regardless of geographic location or financial capacity. Course prices can be adjusted to suit individual affordability, ensuring that professional development opportunities are inclusive and aligned with Sustainable Development Goal 10 (Reduced Inequalities).

Personal Sustainability

Beyond benefiting teachers, the solution encourages Discentis to adopt sustainable practices as a core principle, reinforcing a commitment to long-term social responsibility. It acts as a practical reminder that organizational decisions should align with SDGs, creating enduring value for both individuals and institutions.

5. Feasibility & Next Steps

This section assesses how realistically the EquiLearn design process and solution can be implemented in real-world conditions. The evaluation draws on validated prototypes, stakeholder input from Discentis, and the technical architecture developed in the solution phase.

5.1 Process Feasibility

The design process for EquiLearn is highly replicable and practical for other teams addressing similar educational equity challenges. By combining secondary research from OECD TALIS 2024, primary stakeholder interviews, AI-assisted dataset simulation, and iterative prototyping, the methodology ensures that each stage of development is evidence-based and user-centered.

Strengths:

- **Data-Driven Approach:** Publicly available TALIS 2024 data allowed simulation of teacher participation patterns across Italian regions, providing a solid foundation for testing hypotheses even in the absence of full internal data.
- **Stakeholder Engagement:** Interviews with Discentis validated key gaps—geographic invisibility, lack of teacher interaction, and voluntary dropout rates—ensuring the solution directly addresses real operational challenges.
- **Prototype Validation:** Interactive mockups for the Teacher Dashboard, Equity Dashboard, and AI Teaching Companion allowed early testing of assumptions with measurable success indicators such as recommendation click-throughs, dashboard alert responses, and AI companion usage rates.
- **AI-Assisted Research:** ChatGPT accelerated dataset generation and hypothesis testing, demonstrating a practical method for handling large reports and cross-referencing literature.

Limitations and Mitigations:

The main limitation is reliance on simulated data rather than actual Discentis enrollment records. This is mitigated by grounding simulations in authoritative OECD statistics and confirming assumptions with stakeholders. In a full implementation, real data would replace simulations, strengthening evidence while maintaining the same methodological framework.

5.2 Technical Feasibility

EquiLearn can be realistically developed using the tools, expertise, and resources available to Discentis within 6–12 months. The solution leverages mature, widely used technologies and standard web infrastructure.

Technology Readiness:

- **AI Recommendation Engine:** Uses hybrid collaborative and content-based filtering (Python, scikit-learn), following best practices from platforms like Coursera and LinkedIn Learning.
- **Geographic Equity Dashboard:** Real-time visualizations, heatmaps, and alerts use proven frameworks for data aggregation and analytics.

- AI Teaching Companion: Integrates OpenAI GPT API, delivering responses in under 5 seconds, with standard human moderation for flagged content.
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Web Platform & Payment Integration: Standard cloud-based architecture and Stripe API enable secure, scalable operations.

Resource Requirements:

- Development: 2–3 web developers + 1 ML engineer over 6–12 months.
- Infrastructure: Cloud hosting, OpenAI API, Stripe API.
- Data: Teacher profiles, course catalog metadata, optional postal codes with consent.
- Organizational: Partnership with Discentis, Ministry endorsement, GDPR compliance.
- Skills: Web development, machine learning, Italian language content moderation.

Scalability:

The platform is designed to scale from 5,000 initial users to 50,000+ through modular architecture and cloud-based horizontal scaling, without major redesign.

Implementation Risks and Mitigation:

- Cold Start Problem: Initially limited data for AI recommendations mitigated through content-based matching; collaborative filtering improves as usage data accumulates.
- Partner Dependency: Relies on Discentis maintaining catalogs and responding to enrollments; mitigated via service-level agreements and automated reminders.
- User Engagement: Sustained teacher participation may decline after initial adoption; mitigated through gamification, regular content updates, and community incentives.

5.3 Next Steps

The roadmap below outlines the critical steps to transition EquiLearn from validated prototypes to a fully operational platform. The phased approach is designed to reduce risk, ensure iterative improvements, and maintain momentum toward addressing inequities in teacher training across Italy.

Phase 1: Prototype Validation (Months 1–2)

Conduct live testing with 50–100 teachers across Northern, Central, and Southern regions to validate the three core hypotheses.

Track key success metrics: at least 70% click-through on AI course recommendations, 60% of equity alerts resulting in action within two weeks, and 40% engagement with the AI companion in the first week.

Gather qualitative feedback through interviews to identify usability issues, prioritize features, and refine the user experience.

Share findings with Discentis leadership to confirm partnership terms for pilot development.

Phase 2: Pilot Development (Months 3–8)

Assemble a development team comprising 2–3 web developers and 1 ML engineer with experience in educational technology.

Build a minimum viable product (MVP) including: teacher registration and profiling, AI recommendation engine with content-based matching, equity dashboard for regional tracking, AI teaching companion via OpenAI integration, and a basic community space for discussion.

Integrate with Discentis systems for course catalog updates, enrollment notifications, and completion tracking.

Ensure GDPR compliance with privacy policies, consent management, and anonymized data for equity analytics.

Phase 3: Pilot Launch (Months 9–12)

Launch the pilot with 5,000 teachers from priority regions (Calabria, Sicilia, Campania), where participation gaps are largest.

Monitor performance indicators including regional registration, course completion, AI companion usage, community engagement, and dropout rates.

Iterate rapidly based on user feedback, using bi-weekly review cycles to resolve issues and implement improvements.

Engage the Ministry of Education to demonstrate the equity dashboard's value and secure endorsement for a nationwide rollout.

Phase 4: Scale and Sustainability (Year 2+)

Expand to 50,000+ teachers nationwide, with targeted outreach to underserved regions.

Introduce advanced features such as multilingual support for future international expansion, offline access for low-connectivity areas, and integration with school LMS platforms like Moodle and Microsoft Teams.

Transition from pilot phase (where Discentis retains 100% of fees) to a sustainable revenue-sharing model to fund platform operations.

Measure impact against SDG 4 objectives, including regional participation gap reductions (target $\geq 20\%$ within 18 months), improvements in teacher confidence, and correlations with student learning outcomes.

Feasibility Assessment

EquiLearn demonstrates strong feasibility in both process and technical dimensions. The design

methodology is reproducible, evidence-based, and grounded in validated stakeholder needs. The technical solution leverages established technologies and realistic resources, while prototypes provide a low-risk mechanism to validate assumptions before major development investments.

Key Risks and Mitigations

The main risks are partner dependency and sustaining teacher engagement. These are mitigated through clear partnership agreements, automated workflows, gamification, and proactive community management. With Discentis committed to addressing their three confirmed gaps—geographic visibility, teacher interaction, and dropout reduction—the organizational conditions are favorable for success.

By following this phased roadmap, EquiLearn can move from concept to full operational platform within 12 months, enabling equitable access to high-quality professional development for teachers across Italy and supporting national progress toward SDG 4.